

By defn, column 1 is the pivot column. $\therefore x_1$ is a basic variable
 Since there are no pivots in Column 2 and 3, x_2 and x_3 are free variables.

The corresponding system of equations are: $x_1 - 0.3x_2 - 0.2x_3 = 0.7$

$$\Rightarrow \boxed{x_1 = 0.7 + 0.3x_2 + 0.2x_3}$$

In parametric vector form: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0.3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0.2 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.7 \\ 0 \\ 0 \end{bmatrix}$

+ $\begin{bmatrix} 0.7 \\ 0 \\ 0 \\ p \end{bmatrix}$ The solution set of the nonhomogeneous equation $Ax=b$ is the translated plane $p + \text{Span}\{u, v\}$, which passes through p and is parallel to the solution set of the homogeneous equation in Example 2.

3. Prove the first part of Theorem 6: Suppose that p is a solution of $Ax=b$, so that $Ap=b$. Let v_h be any solution to the homogeneous equation $Ax=0$, and let $w=p+v_h$. Show that w is a solution to $Ax=b$.

Solution: $Aw = A(p+v_h) = Ap + Av_h$ (Thm 5(a))
 $= b + 0$ ($\because Ap=b$ and v_h being a soln. of $Ax=0$)
 $= b$ (Algebraic Property iii)

$\therefore w$ is a solution to $Ax=b$ $\square$

Exercise 1.5

In Exercises 1-4, determine if the system has a nontrivial solution. Try to use as few row operations as possible.

$$\begin{aligned} 1. \quad & 2x_1 - 5x_2 + 8x_3 = 0 \\ & -2x_1 - 7x_2 + x_3 = 0 \\ & 4x_1 + 2x_2 + 7x_3 = 0 \end{aligned}$$

Solution: $x_1 \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ -7 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 8 \\ 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

By defn. of $Ax=b$, we get: $\begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. This equation has a non-trivial solution if and only if the equation has at least one free variable.